

量子 2018

$$(1) -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi = E\psi$$

$$\psi_2 = \psi \cdot e^{ia}$$

$$-\frac{\hbar^2}{2m} \nabla^2 (e^{ia}\psi) + V(e^{ia}\psi) = E(e^{ia}\psi)$$

$$\Rightarrow -\frac{\hbar^2}{2m} \nabla^2 \psi_2 + V\psi_2 = E\psi_2$$

ψ_2 也是这个体系的可能函数。

线性关系，是同一本征值的本征态。

(2) 一般的态 $\psi = \cos\theta\psi_1 + \sin\theta\psi_2$ 。

$$L\psi_1 = E_1\psi_1, \quad L\psi_2 = E_2\psi_2$$

体系可能值为 E_1 ，概率 $|\cos\theta|^2$

E_2 ， $|\sin\theta|^2$

依据是体系的可能值是力学量算符的本征值。

$$(3) \psi(x, t) = \sum_n c_n e^{-iE_n t/\hbar} \psi_n$$

$$\text{其中 } c_n = \langle \psi_n | \psi(x, 0) \rangle$$

二、假设自旋 $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$

玻色子，交换对称

$$\psi = \psi_m(x_1)\psi_n(x_2) \begin{cases} |\uparrow\uparrow\rangle \\ \frac{1}{\sqrt{2}}[|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle] \\ |\downarrow\downarrow\rangle \end{cases}$$

$$\psi = \frac{1}{\sqrt{2}} [\psi_m(x_1)\psi_n(x_2) + \psi_m(x_2)\psi_n(x_1)]$$

$(m \neq n)$

$$\begin{cases} |\uparrow\uparrow\rangle \\ \frac{1}{\sqrt{2}}[|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle] \\ |\downarrow\downarrow\rangle \end{cases}$$

$$\psi = \frac{1}{2} [\psi_m(x_1)\psi_n(x_2) - \psi_m(x_2)\psi_n(x_1)] [|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle]$$

费米子，交换反对称

$$\psi = \psi_m(x_1)\psi_n(x_2) \cdot \frac{1}{\sqrt{2}} [|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle]$$

$$\psi = \frac{1}{2} [\psi_m(x_1)\psi_n(x_2) + \psi_m(x_2)\psi_n(x_1)] [|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle]$$

$$\psi = \frac{1}{\sqrt{2}} [\psi_m(x_1)\psi_n(x_2) - \psi_m(x_2)\psi_n(x_1)] \begin{cases} |\uparrow\uparrow\rangle \\ \frac{1}{\sqrt{2}}[|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle] \\ |\downarrow\downarrow\rangle \end{cases}$$

三、球谐函数 $Y_{11} = -\frac{1}{2}\sqrt{\frac{3}{2\pi}} e^{i\varphi} \sin\theta = -\frac{1}{2}\sqrt{\frac{3}{2\pi}} \frac{x+iy}{r}$

$Y_{1-1} = \frac{1}{2}\sqrt{\frac{3}{2\pi}} e^{-i\varphi} \sin\theta = \frac{1}{2}\sqrt{\frac{3}{2\pi}} \frac{x-iy}{r}$

$Y_{10} = \frac{1}{2}\sqrt{\frac{3}{\pi}} \cos\theta = \frac{1}{2}\sqrt{\frac{3}{\pi}} \frac{z}{r}$

1) $\psi = A(x+y+z)e^{-\alpha r}$

$= A \left[\sqrt{\frac{2\pi}{3}} (Y_{1-1} - Y_{11}) + i\sqrt{\frac{2\pi}{3}} (Y_{1-1} + Y_{11}) + 4\sqrt{\frac{\pi}{3}} Y_{10} \right] r e^{-\alpha r}$

~~$= A \sqrt{\frac{2\pi}{3}} (1+i) Y_{11} r e^{-\alpha r} + A \sqrt{\frac{2\pi}{3}} (i-1) Y_{1-1} r e^{-\alpha r}$~~
 ~~$+ 2A \sqrt{\frac{\pi}{3}} Y_{10} r e^{-\alpha r}$~~

$= \sqrt{\frac{2\pi}{3}} A r e^{-\alpha r} [(1+i) Y_{11} + (i-1) Y_{1-1} + 2\sqrt{2} Y_{10}]$

$l=1, L^2 \psi = 2\hbar^2 \psi$

$\Rightarrow \langle L^2 \rangle = 2\hbar^2$

(2) $\psi = A \sqrt{\frac{2\pi}{3}} r e^{-\alpha r} [(1+i) Y_{11} + (i-1) Y_{1-1} + 2\sqrt{2} Y_{10}]$

归一化有 $A = \frac{1}{\sqrt{8\pi}}$

$\psi = R(r) \left[\frac{1+i}{\sqrt{2}} Y_{11} + \frac{i-1}{\sqrt{2}} Y_{1-1} + \sqrt{\frac{2}{3}} Y_{10} \right]$

$L_z Y_{lm} = m\hbar Y_{lm}$

$\langle L_z \rangle = \frac{1}{6}\hbar - \frac{1}{6}\hbar = 0$

(3) $L_z Y_{11} = \hbar Y_{11}$

$P = \left| \frac{1+i}{\sqrt{2}} \right|^2 = \frac{1}{6}$

(4) $x = r \sin\theta \cos\varphi$

$y = r \sin\theta \sin\varphi$

$z = r \cos\theta$

$\psi = A (\sin\theta \cos\varphi + \sin\theta \sin\varphi + 2\cos\theta) r e^{-\alpha r}$

$dp = |\psi|^2 d\Omega = |A|^2 (\sin\theta \cos\varphi + \sin\theta \sin\varphi + 2\cos\theta)^2 r^2 e^{-2\alpha r} r^2 \sin\theta dr d\theta d\varphi$

四. (1) 在 S_z 表象中, 有

$$H = A \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + B \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$= \frac{\hbar}{2} \begin{pmatrix} B & -iA \\ iA & -B \end{pmatrix}$$

$$\text{由 } \frac{\hbar}{2} \begin{pmatrix} B & -iA \\ iA & -B \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = E \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\Rightarrow \begin{vmatrix} E - \frac{\hbar}{2}B & \frac{i\hbar}{2}A \\ -\frac{i\hbar}{2}A & E + \frac{\hbar}{2}B \end{vmatrix} = 0$$

$$E = \pm \frac{\hbar}{2} \sqrt{B^2 - A^2}, \text{ 令 } \sqrt{B^2 - A^2} = C$$

$$E = \frac{\hbar}{2} \sqrt{B^2 - A^2} = \frac{\hbar}{2} C, \psi_+ = \frac{1}{\sqrt{2B^2 - 2BC}} \begin{pmatrix} iA \\ B - C \end{pmatrix}$$

$$E = -\frac{\hbar}{2} C, \psi_- = \frac{1}{\sqrt{2B^2 + 2BC}} \begin{pmatrix} iA \\ B + C \end{pmatrix}$$

$$(2) \psi = a\psi_+ + b\psi_-$$

$$\langle S_y \rangle = |a|^2 \langle \psi_+ | S_y | \psi_+ \rangle + |b|^2 \langle \psi_- | S_y | \psi_- \rangle$$

$$= |a|^2 \cdot 0 + |b|^2 \cdot 0$$

$$= 0$$

$$(3) S_y = +\frac{\hbar}{2}, \psi = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$P_+ = |\langle \psi | \psi_+ \rangle|^2 = \frac{(A+B-C)^2}{4(B^2-BC)}$$

$$P_- = |\langle \psi | \psi_- \rangle|^2 = \frac{(A+B+C)^2}{4(B^2+BC)}$$

$$\text{五. (1)} \hat{L}_z = -i\hbar \frac{d}{d\varphi}$$

$$H = -\frac{\hbar^2}{2I} \frac{d^2}{d\varphi^2}$$

$$\text{由 } \frac{d^2}{d\varphi^2} \psi + \frac{2IE}{\hbar^2} \psi = 0$$

$$\Rightarrow \psi(\varphi) = A e^{im\varphi}, m = \sqrt{\frac{2IE}{\hbar^2}}$$

$$\text{归一化, } A = \frac{1}{\sqrt{2\pi}}$$

$$\psi(0) = \psi(2\pi)$$

$$\Rightarrow e^{im \cdot 2\pi} = 1 \Rightarrow m = 0, \pm 1, \pm 2, \dots$$

$$\psi(\varphi) = \frac{1}{\sqrt{2\pi}} e^{im\varphi}, m = 0, \pm 1, \pm 2, \dots$$

$$(2) H' = A \sin(2\varphi)$$

$$\text{基态 } \psi_0 = \frac{1}{\sqrt{2\pi}}$$

$$\text{一阶微扰 } E_0^{(1)} = \langle \psi_0 | H' | \psi_0 \rangle$$

$$= \frac{1}{2\pi} \int_0^{2\pi} A \sin 2\varphi d\varphi = 0$$

$$\text{第一激发态, } E_1^{(0)} = \frac{\hbar^2}{2I}$$

$$\psi_1^{(0)} = \frac{1}{\sqrt{2\pi}} e^{i\varphi}, \psi_{-1}^{(0)} = \frac{1}{\sqrt{2\pi}} e^{-i\varphi}$$

二重简并

$$H'_{11} = \langle \psi_1^{(0)} | H' | \psi_1^{(0)} \rangle = \frac{1}{2\pi} \int_0^{2\pi} A \sin 2\varphi d\varphi = 0$$

$$H'_{1,-1} = \langle \psi_1^{(0)} | H' | \psi_{-1}^{(0)} \rangle = \frac{1}{2\pi} \int_0^{2\pi} A e^{-2i\varphi} \sin 2\varphi d\varphi$$

$$= -\frac{i}{2} A$$

$$H'_{-1,1} = H'_{1,-1}^* = \frac{i}{2} A$$

$$H'_{-1,-1} = \langle \psi_{-1}^{(0)} | H' | \psi_{-1}^{(0)} \rangle = \frac{1}{2\pi} \int_0^{2\pi} A \sin 2\varphi d\varphi = 0$$

$$H' = \begin{pmatrix} 0 & -\frac{i}{2}A \\ \frac{i}{2}A & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -\frac{i}{2}A \\ \frac{i}{2}A & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = E_1^{(1)} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\begin{vmatrix} E_1^{(1)} & -\frac{i}{2}A \\ -\frac{i}{2}A & E_1^{(1)} \end{vmatrix} = 0$$

$$\Rightarrow E_1^{(1)} = \pm \frac{A}{2}$$

$$S_y = \frac{\hbar}{2}, \quad \psi_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$S_y = -\frac{\hbar}{2}, \quad \psi_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$P(S_y = \frac{\hbar}{2}) = |\langle \psi_+ | \psi(t) \rangle|^2$$

$$= \frac{1}{4} \left| (1, i) \begin{pmatrix} e^{-i\frac{eBt}{4m}} \\ e^{i\frac{eBt}{4m}} \end{pmatrix} \right|^2$$

$$= \frac{1}{4} \times 2 \left(\cos \frac{eBt}{4m} - \sin \frac{eBt}{4m} \right)^2$$

$$= \frac{1}{2} - \frac{1}{4} \sin \frac{eBt}{2m}$$

$$P(S_y = -\frac{\hbar}{2}) = |\langle \psi_- | \psi(t) \rangle|^2$$

$$= \frac{1}{4} \left| (1, -i) \begin{pmatrix} e^{-i\frac{eBt}{4m}} \\ e^{i\frac{eBt}{4m}} \end{pmatrix} \right|^2$$

$$= \frac{1}{2} + \frac{1}{4} \sin \frac{eBt}{2m}$$

六. (1) 在 S_x 表象中, $S_x = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$H = \frac{eB\hbar}{4m} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$E_+ = \frac{eB\hbar}{4m}, \quad \psi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$E_- = -\frac{eB\hbar}{4m}, \quad \psi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$t=0 \text{ 时, } S_y \text{ 为正, } \psi_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{2}} \psi_+ + \frac{1}{\sqrt{2}} \psi_-$$

$$\psi(t) = \frac{1}{\sqrt{2}} e^{-iE_+t/\hbar} \psi_+ + \frac{1}{\sqrt{2}} e^{-iE_-t/\hbar} \psi_-$$

$$= \frac{1}{\sqrt{2}} e^{-i\frac{eBt}{4m}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} e^{i\frac{eBt}{4m}} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-i\frac{eBt}{4m}} \\ e^{i\frac{eBt}{4m}} \end{pmatrix}$$

(2) S_x 表象中

$$S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$$